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PLANETARY LANDER GAME

An interactive educational game modelled after the seven minutes of Terror during the Martian approach, We aim to make the game as educational as possible by adding in real-life information from different landing missions.

Our primary goal is to have the player follow through all the important stages of approach and descent while simulating just a fraction of the high stakes poker game that takes place during real-world missions.

CURIOSITY

7 MINUTES OF TERROR

Approaching Mars at
13,000
mph (estimate)

To enable a successful entry, descent, and landing, engineers design a series of operations during the approach phase. At days before the spacecraft enters the Martian atmosphere.

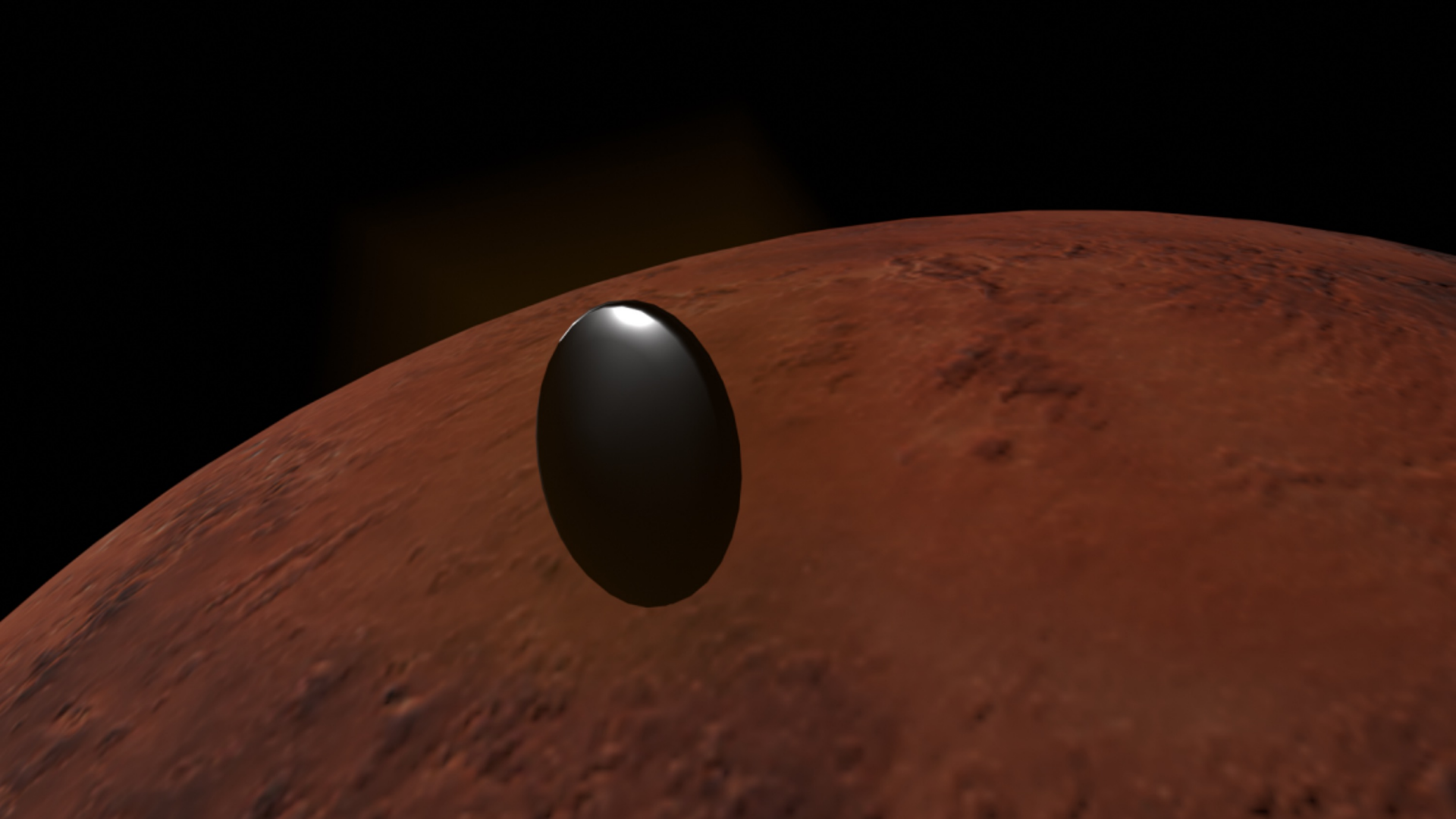
The Deep Space Network's 24 major and 13 minor antennas will be used to provide real-time coverage of the spacecraft during the approach phase.

GETTING FROM THE TOP OF THE ATMOSPHERE DOWN TO THE SURFACE TAKES 7 MINUTES

CURIOSITY CARRIES OVER 10 TIMES THE MASS OF SCIENTIFIC INSTRUMENTS AS THE SPIRIT OR OPPORTUNITY ROVERS

ENTRY







Objective

Survive 10 minutes of Terror

The stakes:

\$3 Billion Dollars

1000+ Man hours

The hopes and dreams

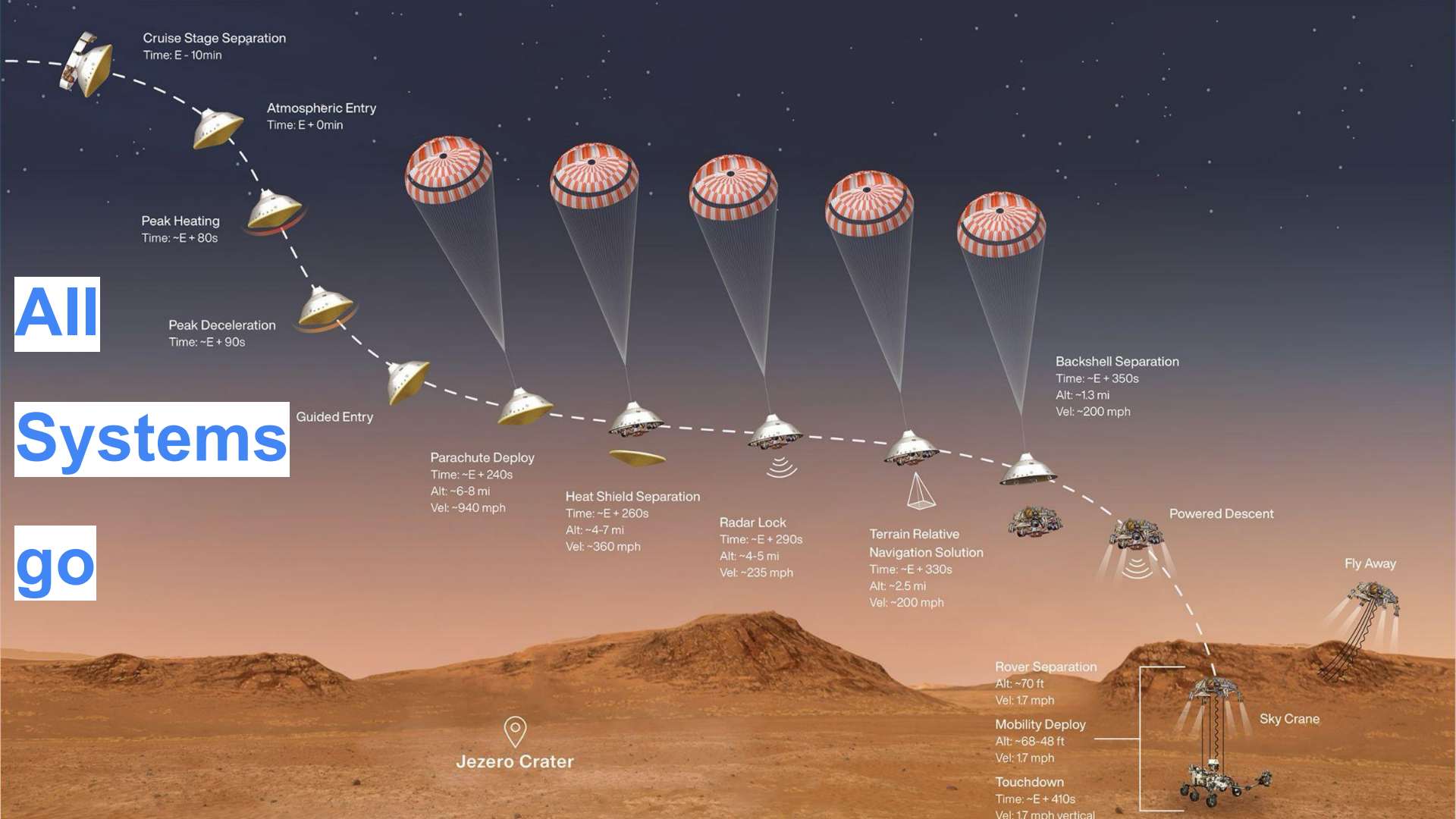
Of millions.

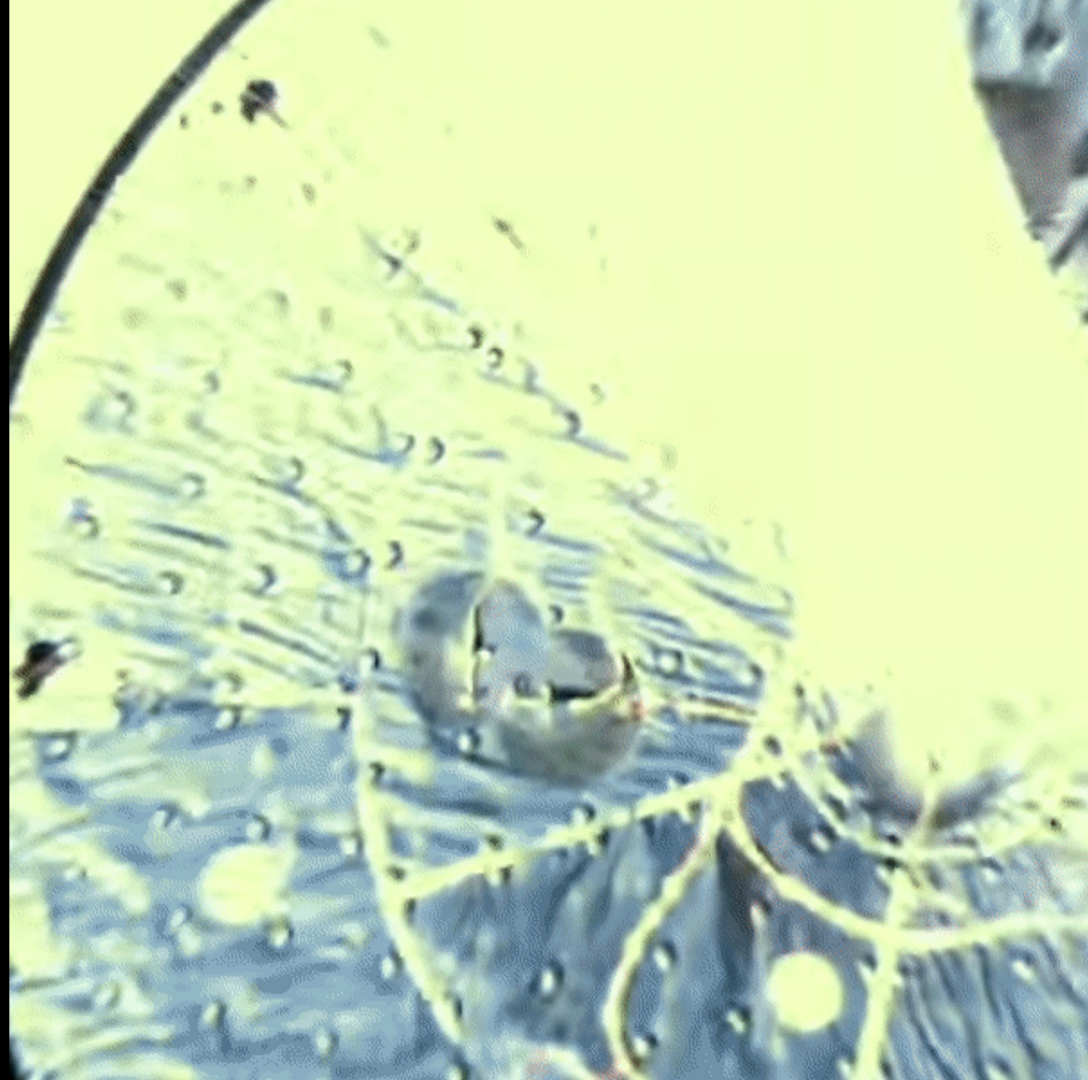


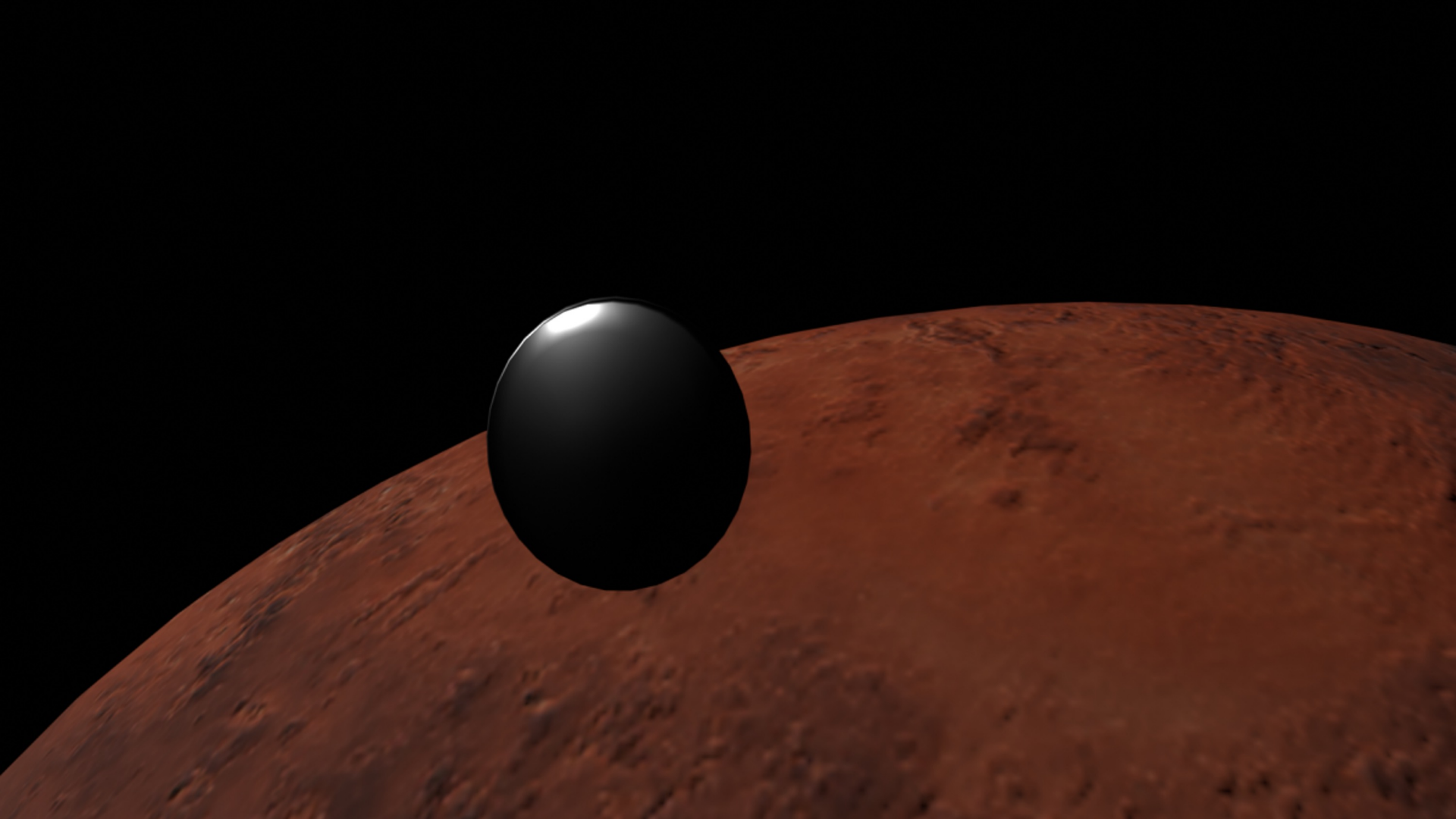
All

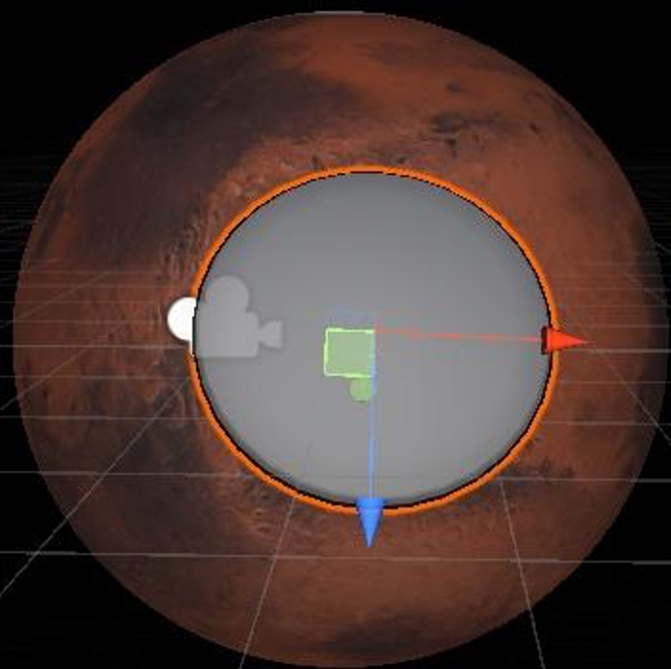
Systems

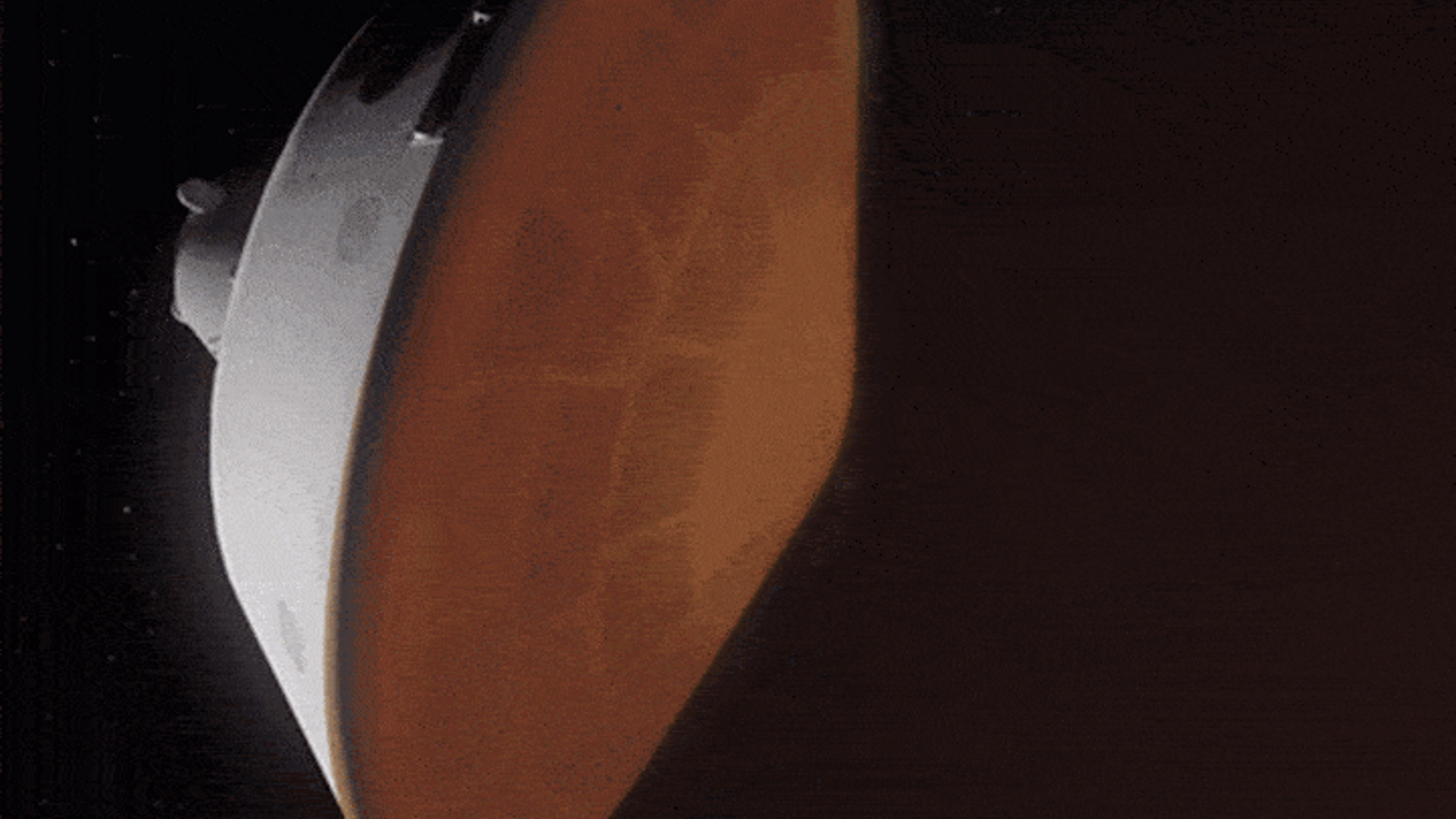
go









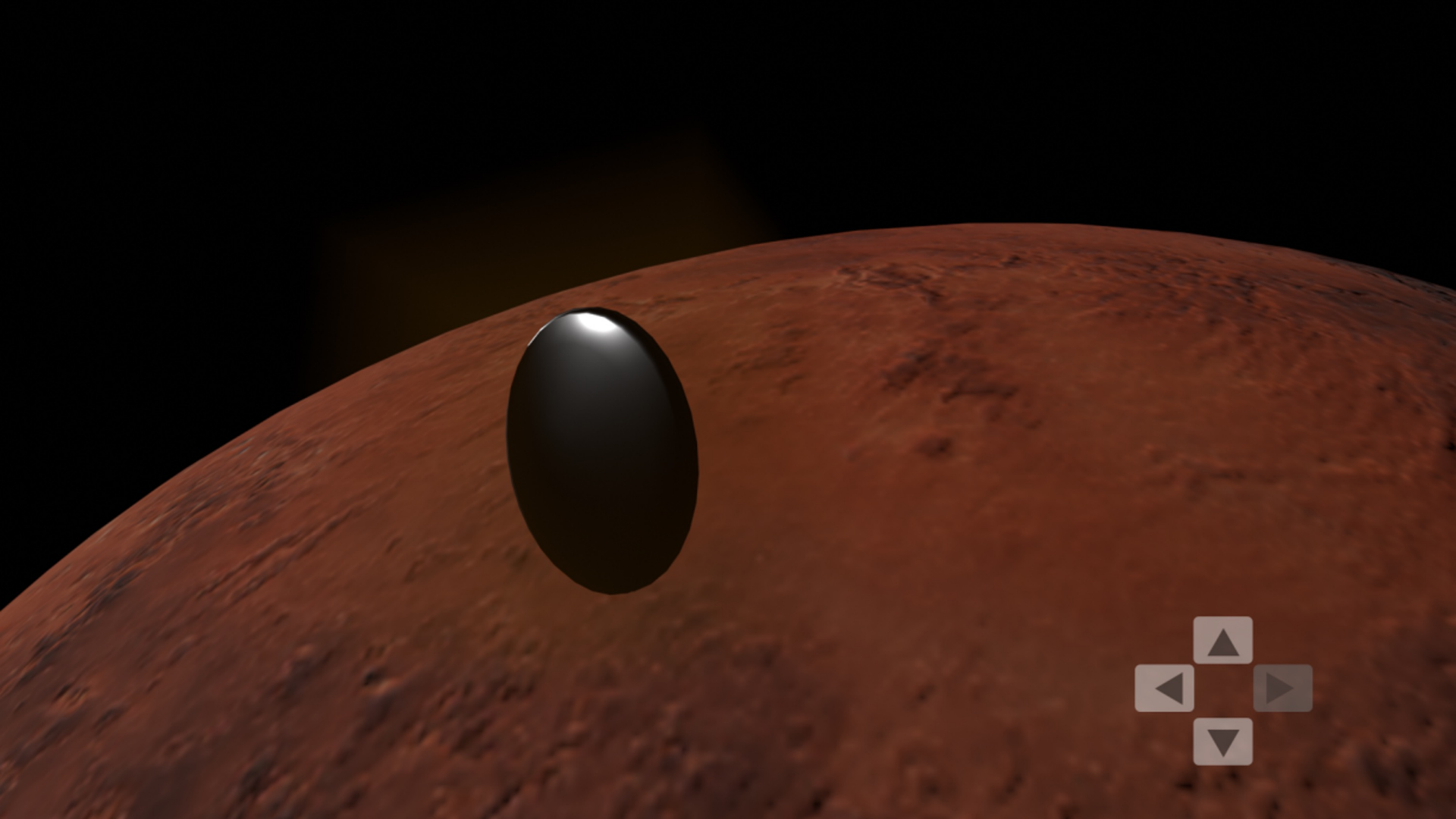


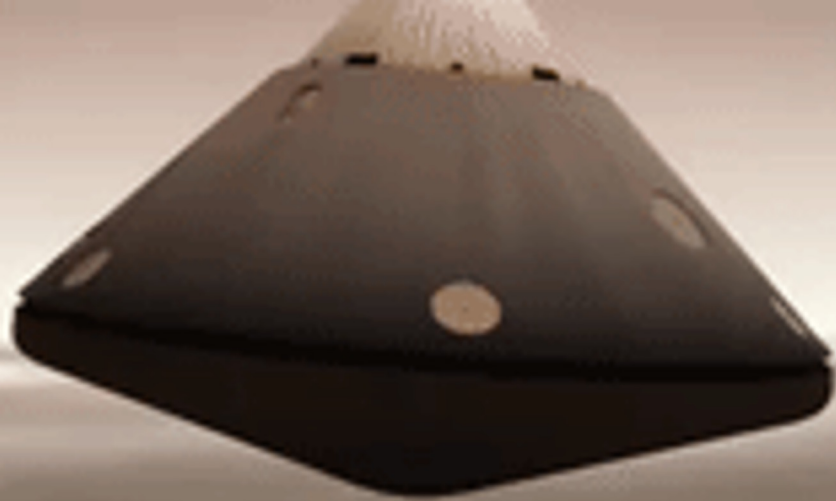
30% speed



real time

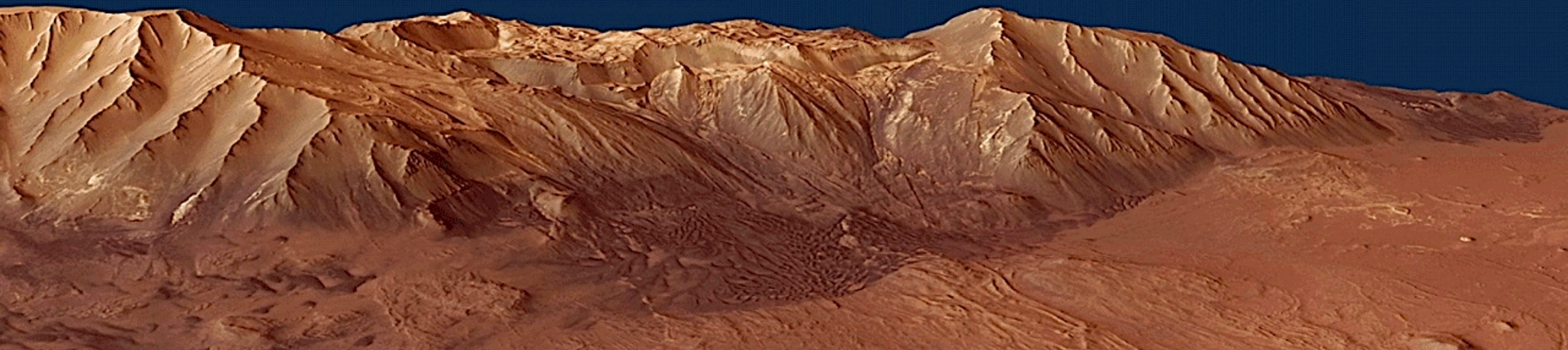






MARS 2020 ROVER

TERRAIN RELATIVE NAVIGATION



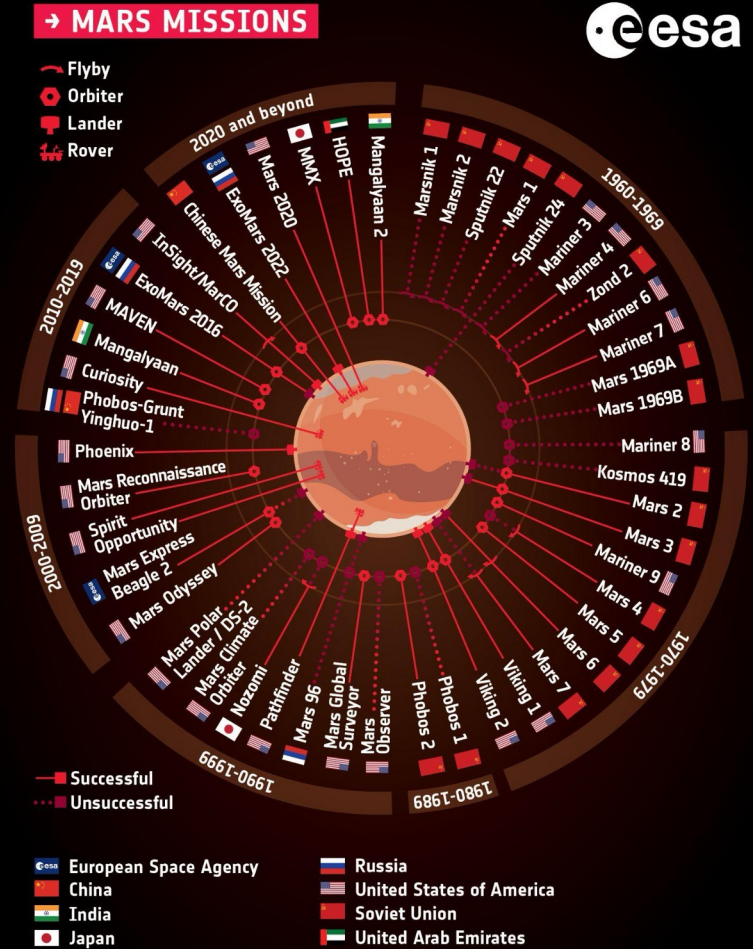




Task failed successfully.

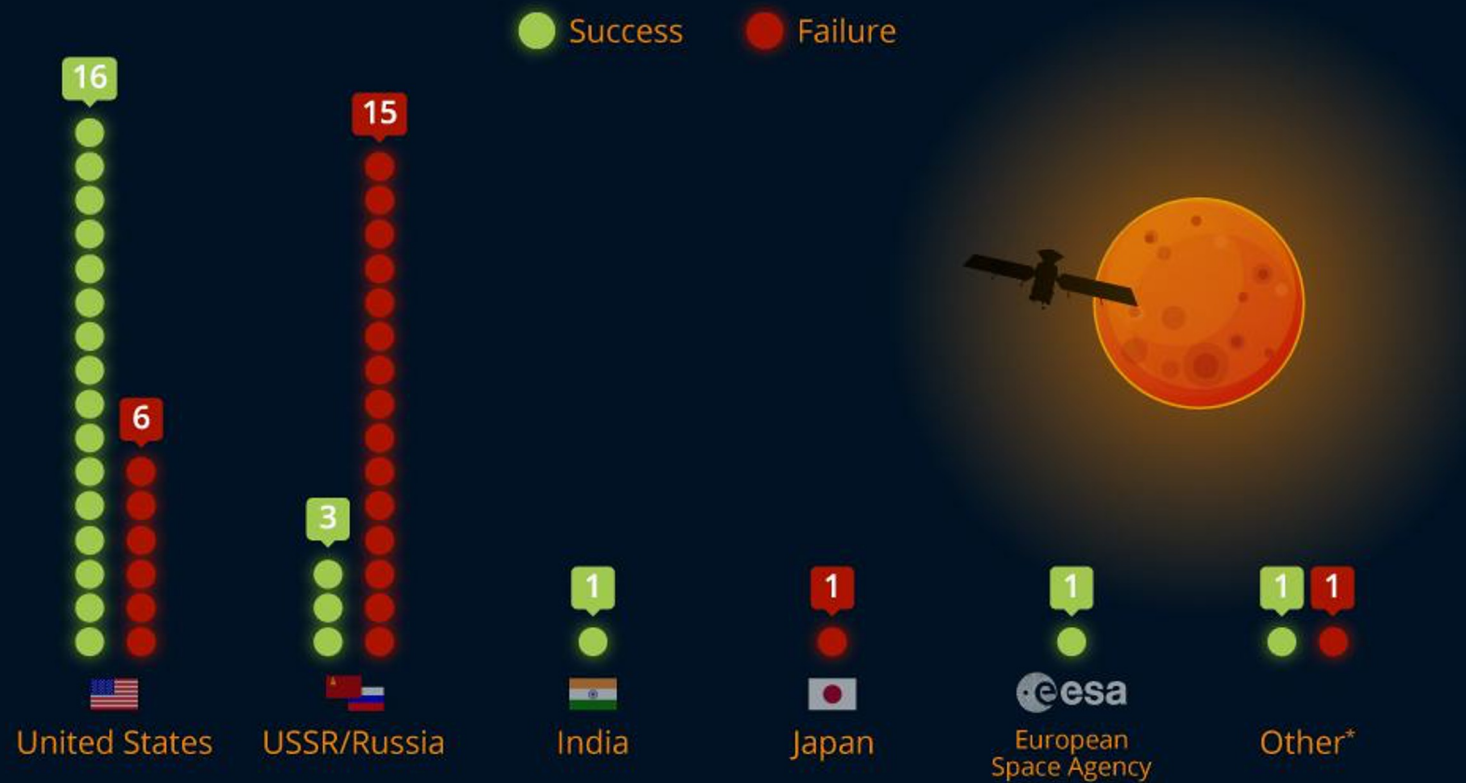
Those who do not learn from history are destined to repeat their mistakes.

As Part of our game everytime the player fails , we will give them a short text about real life missions which failed under similar circumstances and how the respective agencies learnt from them to better their subsequent missions.



Mars: Man's Enduring Curiosity

Number of successful/unsuccessful unmanned missions to Mars since 1960



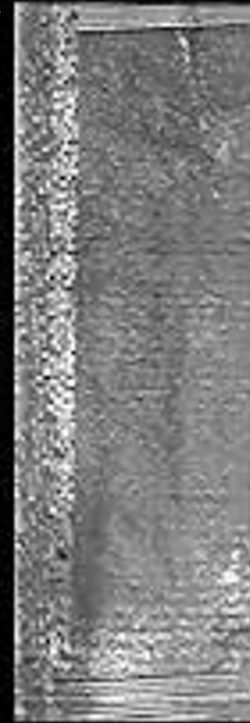
* Other refers to joint Russia/China and ESA/Russia missions.
The former was a failure and the latter was a partial success.

Expanded and Enhanced

-> Historical missions simulations, to illustrate how advances in technology have changed martian landings.

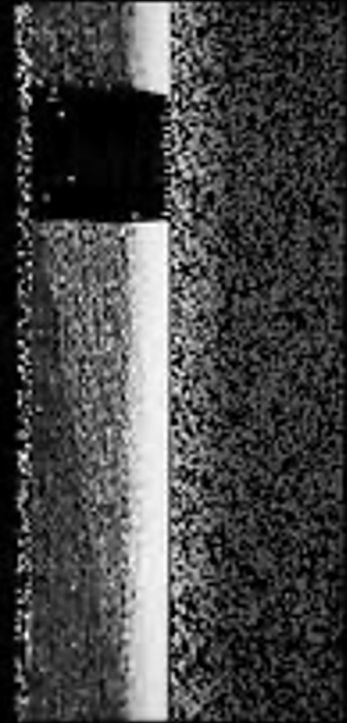
-> Introducing RnG elements to simulate Certain fail states like Radio failure ,fuel leaks and module disintegration during descent to name a few.

->implement new game mechanics to illustrate the degree of minor corrections undertaken by the onboard systems to ensure successful landing.



20 sec. Signal

14.5 sec. Video



Detail of digital telemetry and video

Video appears to be pure noise

D.P.M.